

From Predictive Analytics to Operational Decision Support

A practical framework for NovaBank customer outreach prioritization

Prepared by

Neelima Parupally

**Analytics Methods and Frameworks Project
Quantic School of Business and Technology**

July 2026

Introduction: the decision to support

A genuine, operational framing of the NovaBank case

BUSINESS PROBLEM

NovaBank needs to use limited retention/campaign capacity more effectively. The goal is not to contact everyone; it is to prioritize customers most likely to respond to proactive outreach.

ANALYTICS TRANSLATION

- Binary classification: estimate probability of a positive customer response
- Ranking problem: decide who is in the top outreach group.
- Decision framework: connect model scores to action tiers and pilot governance

IMPORTANT LIMITATION

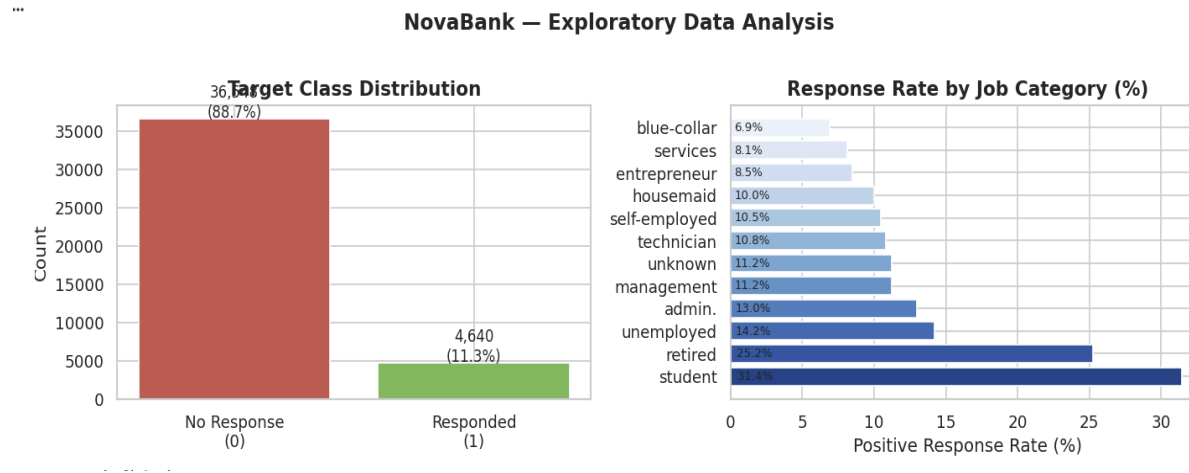
The public dataset measures term-deposit campaign response, not actual churn. I use response as a proxy for retention engagement and clearly treat financial impact as an illustrative scenario.

SUCCESS MEASURES

- AUC-ROC: ranking quality across all customers
- Average precision / Precision@Top20%: usefulness under class imbalance
- Pilot lift vs. control: the final business proof

Data readiness: response is rare and uneven

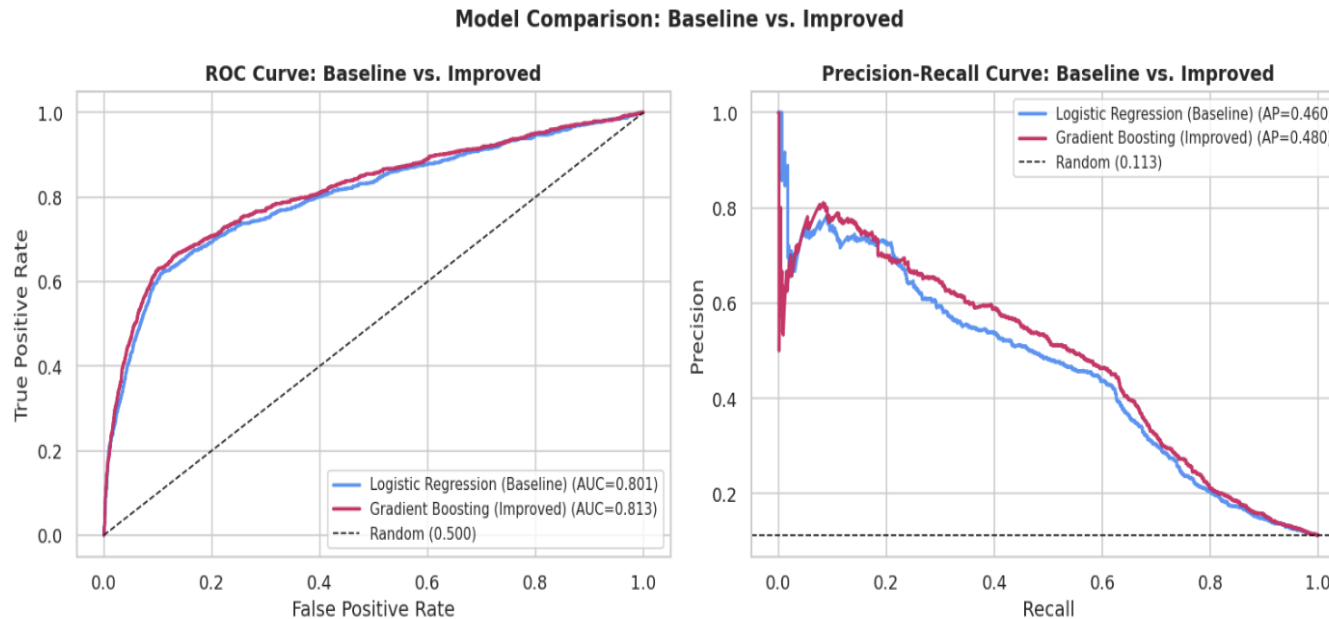
UCI Bank Marketing dataset used as a proxy for NovaBank outreach response



- 41,188 records with 20 input features plus a binary target
- Only 11.3% positive response, so accuracy alone is not useful
- Duration is excluded to prevent leakage because it is only known after a call
- Macroeconomic and prior-contact variables show stronger signal than demographics alone

Modeling approach: baseline first, then improvement

Logistic regression establishes the floor; gradient boosting improves ranking



BASELINE

- Logistic regression with class weighting
- Easy to explain and audit
- Baseline AUC: 0.801; AP: 0.460

IMPROVED MODEL

- Gradient boosting captures non-linear patterns
- Improved AUC: 0.814; AP: 0.489
- Gain is modest but useful for prioritizing thousands of customers

3x+

estimated lift in top-ranked
group vs. random

Top 20%

recommended capacity-
based outreach pool

Explainability: what drives the ranking?

Permutation importance shows predictive contribution, not causation

WHAT LEADERSHIP CAN UNDERSTAND

- Economic context matters: employment and interest-rate indicators are strong drivers
- Channel, timing, and prior campaign outcome affect customer response
- Demographics are not the primary standalone drivers and should be monitored for fairness

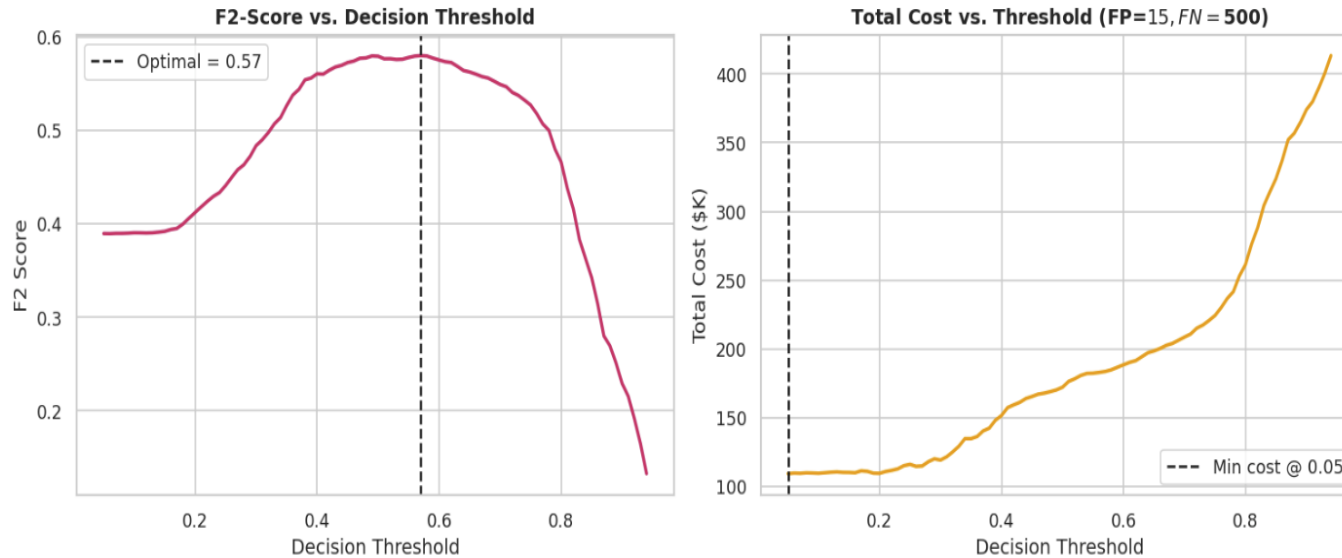
GOVERNANCE NOTE

This model should be used as a ranking aid, not an automatic decision engine. Final production use requires monitoring, fairness review, and business-owner signoff.

Decision framework: use capacity first, thresholds second

Avoid mixing three different decision rules

Threshold Optimization — Decision Framework



ACTION TIERS

Tier 1: highest scores, personal call

Tier 2: next-highest scores, email + follow-up

No action: hold this cycle

PRIMARY PRODUCTION RULE

Rank customers by predicted probability and contact the top 20%, because the quarterly campaign has limited capacity.

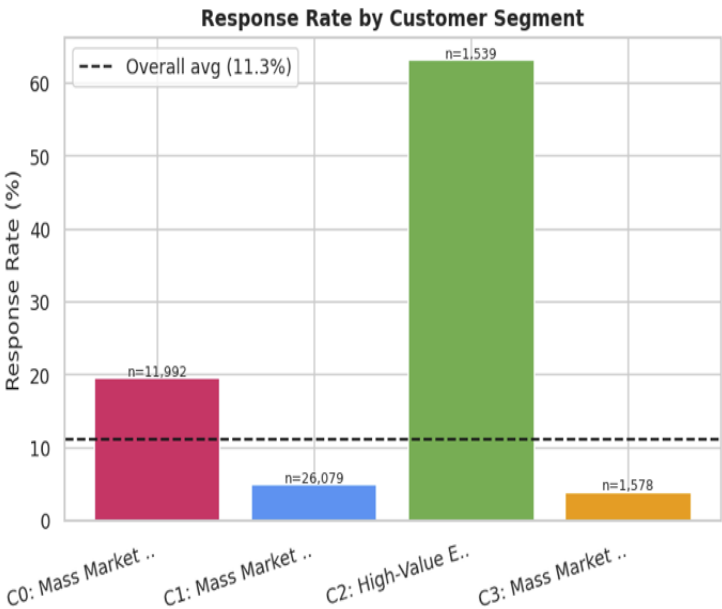
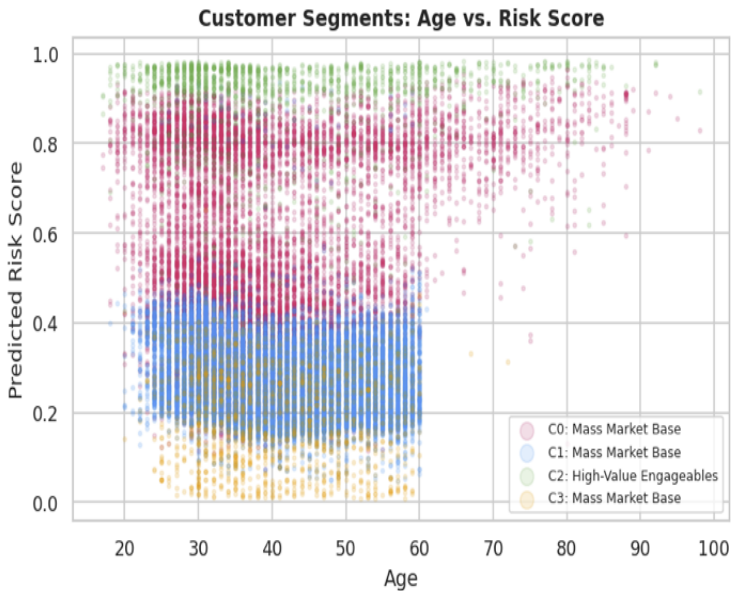
SUPPORTING THRESHOLD ANALYSIS

- F2-score peaks around 0.46 when recall is valued more than precision
- Minimum-cost threshold is lower under the assumed \$15 FP / \$500 FN cost ratio
- These thresholds are sensitivity checks, not replacements for the top-20% capacity rule

Segmentation: useful for messaging, not proof of model performance

K-means highlights one high-response group and several lower-response groups

K-Means Customer Segmentation (k=4)



CORRECTED INTERPRETATION

- The segment chart does not support four sharply distinct customer personas
- It supports a simpler message: one cluster has much higher response, while the other groups are lower-response mass-market segments
- Segment labels should be finalized only after reviewing cluster profiles and sizes

HOW TO USE THIS

Use segmentation to tailor outreach scripts and offers. Do not use it as independent proof that the predictive model works; performance must be validated on the test set and pilot control group.

Validation plan: prove lift, fairness, and operational fit

The model should not go directly into full production

90-DAY PILOT DESIGN

- 5,000 customers total: 2,500 model-selected and 2,500 randomized control
- Measure incremental response / retention lift, cost per successful contact, and customer complaints
- Track results by age group and job category for fairness review

FAIRNESS CHECKS TO INCLUDE

- Selection rate by group
- Precision by group among contacted customers
- Recall / missed-positive rate by group
- Escalation review if gaps are large or sample sizes are too small

RISKS CONTROLLED

- Proxy-target risk: campaign response is not actual churn
- Leakage risk: call duration is removed
- Drift risk: macro indicators can change
- Customer-fatigue risk: outreach volume must be capped

GO / NO-GO CRITERIA

Scale only if the pilot shows commercially meaningful lift vs. control, acceptable fairness metrics, and manageable Tier 1 call volume.

Recommendation: pilot the ranking model, then scale carefully

A clear decision, with transparent assumptions

RECOMMENDED DECISION

Approve a 90-day pilot using the gradient boosting score to rank customers. Contact the highest-scored 20% using two outreach tiers, while holding out a randomized control group.

WHY THIS IS CREDIBLE

- Baseline and improved model are both shown
- Decision threshold is tied to business capacity and cost sensitivity
- Unsupported ROI claims are avoided; financial impact is treated as a pilot-tested scenario
- Reproducible notebook and dataset are included for review

WHAT LEADERSHIP GETS

- A prioritized outreach list
- A defensible reason for who gets contacted first
- A test plan before spending the full budget
- A governance path for fairness and drift monitoring

FINAL ANSWER

Use the model to prioritize action, not to automate judgment.

References and AI usage transparency

Sources and reproducibility notes

DATASET AND CITATION

Moro, S., Cortez, P., & Rita, P. (2014). A data-driven approach to predict the success of bank telemarketing. Decision Support Systems. UCI Bank Marketing dataset, bank-additional-full.csv.

AI USAGE LOG

- AI was used to improve narrative structure, identify gaps, and polish executive wording
- Analysis assumptions, code review, and final recommendations were checked and owned by the analyst
- AI-generated suggestions were revised to avoid unsupported claims and to align with the project rubric

FINAL SUBMISSION LINKS

GitHub notebook link:

[https://github.com/neelimaparupally/Novabank_Analytics_Project/blob/main/novabank_analysis_submission_ready.ipynb]